

Divisibility Rules

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Introduction

Let's start with what "divisibility" means.

Definition 0.1 – Divisibility If when $\spadesuit$ divided by $\diamondsuit$ does **NOT** leave a remainder, we can say $\spadesuit$ is *divisible* by $\diamondsuit$.

Example 0.1 – divisible $21 \div 7 = 3$, remainder 0.

Here, we see there is no remainder, so we can say that 21 is *divisible* by 7

Example 0.2 – not divisible $65 \div 8 = 8$, remainder 1.

Here we have a remainder. Therefore we call 65 to be *not divisible* by 8.

1 Basic Rules

1.1 Rule for 2

This is the easiest one to spot. If our number ends with 0, 2, 4, 6, 8 (it is an even number), then it's divisible by 2.

1.2 Rule for 3

For this one, we will add up all the digits in a number. If the total is divisible by 3, then the original number also will be.

Example 1.1 Is 852 divisible by 3?

If we add up the digits in 852, $8 + 5 + 2 = 15$. We also know that 15 is divisible by 3, so we can use the rule, and say that yes, 852 is divisible by 3 ($852 = 284 \times 3$).

Example 1.2 Is 412 divisible by 3?

$4 + 1 + 2 = 7$, and 7 is not a multiple of 3. So no, 412 is NOT divisible by 3.

1.3 Rule for 4

For this rule, we look at the last two digits of a number. If those last two digits form a number divisible by 4, then the entire number is divisible by 4.

Example 1.3 Is 3516 divisible by 4?

We look at the last two digits: 16. Since $16 \div 4 = 4$ with no remainder, 3516 is divisible by 4 ($3516 = 879 \times 4$).

Example 1.4 Is 2342 divisible by 4?

The last two digits are 42. We have $42 \div 4 = 10$ remainder 2. Since there is a remainder, 2342 is NOT divisible by 4.

1.4 Rule for 5

This is another easy one. If our number ends with a 5 or a 0, then it is divisible by 5.

1.5 Rule for 6

For 6, we just need to check that our number is divisible by **both** 2 and 3. If both of these are true, then our number is a multiple of 6.

1.6 Rule for 8

For this rule, we look at the last three digits of a number. If those last three digits form a number divisible by 8, then the entire number is divisible by 8. (This is very similar to the rule for 4).

Example 1.5 Is 5432 divisible by 8?

We look at the last three digits: 432. Since $432 \div 8 = 54$ with no remainder, 5432 is divisible by 8 ($5432 = 679 \times 8$).

Example 1.6 Is 3217 divisible by 8?

The last three digits are 217. We have $217 \div 8 = 27$ remainder 1. Since there is a remainder, 3217 is NOT divisible by 8.

1.7 Rule for 9

This one is very similar to the rule for 3. We will add up all the digits in the number, and if the total is divisible by 9, then so will the original number.

Example 1.7 Is 9720 divisible by 9?

$9 + 7 + 2 + 0 = 18$, and 18 is a multiple of 9, so we can say that 9720 is also divisible by 9.

1.8 Rule for 10

If our number ends in a 0, then it is divisible by 10.

Problem 1.1 Can you tell me why?

2 More Advanced

2.1 Rule for 7

For this rule, we use a divisibility test: take the last digit, double it, and subtract it from the rest of the number. If the result is divisible by 7, then the original number is too. Repeat this process if needed.

Example 2.1 Is 343 divisible by 7?

Take the last digit 3, double it to get 6, and subtract from 34: $34 - 6 = 28$. Since $28 \div 7 = 4$ with no remainder, 343 is divisible by 7 ($343 = 49 \times 7$).

2.2 Rule for 11

For this rule, we alternately add and subtract the digits from right to left. If the result is divisible by 11, then the original number is too.

Example 2.2 Is 121 divisible by 11?

Starting from the right: $1 - 2 + 1 = 0$. Since 0 is divisible by 11, we can say that 121 is divisible by 11 ($121 = 11 \times 11$).

Example 2.3 Is 2728 divisible by 11?

Starting from the right: $8 - 2 + 7 - 2 = 11$. Since 11 is divisible by 11, 2728 is divisible by 11 ($2728 = 248 \times 11$).

3 Problems for practice

Problem 3.1 A bakery is packaging cookies into boxes. They have 348 cookies and want to pack them into boxes with 6 cookies each. Will all cookies fit evenly into boxes with no cookies left over?

Problem 3.2 A farmer has some apples. When arranged in rows of 5, there are 3 left over. When arranged in rows of 7, there are 2 left over. What is the least number of apples the farmer could have?

Problem 3.3 A teacher has 285 pencils to distribute equally among her students. If she has 9 students in her class, can she give each student the same number of pencils with none left over?

Problem 3.4 A library is organizing books on shelves. When they try to place books in stacks of 4, they have 1 left over. When they arrange them in stacks of 6, they have 5 left over. What is the minimum number of books they could have?

Problem 3.5 A community garden is planting flowers in rows. They have 810 flower seeds and want to plant them in groups of 9. Will all seeds be used evenly with none remaining?

Problem 3.6 A factory produces widgets that must be packaged in quantities divisible by 2, 3, and 5. What is the smallest number of widgets they need to produce so the packages can be made with no widgets left over?